

The Convex Origin of Fixed Costs¹

Online Appendix

Contents

A Proofs from Section 2 (The duality)	2
A.1 Proof of Proposition 1 (Cost function properties)	2
A.2 Proof of Corollary 1 (Shape of G determines curvature)	2
A.3 Proof of Corollary 2 (Canonical distributions)	2
A.4 Proof of Proposition 2 (Universality)	2
A.5 Topological non-robustness of the degenerate case	3
A.6 Proof of Proposition 3 (Dispersion and convexity)	3
A.7 Proof of Proposition 4 (Entropy correspondence)	3
A.8 Detailed entropy discussion	4
A.9 Proof of Proposition 5 (Multi-action duality)	5
B Investment application	6
C Proofs from the pricing section	8
C.1 Proof of Proposition 7 (Stationary distribution)	9
C.2 Proof of Proposition 8 (Selection and the menu cost distribution)	10
C.3 Proof of Proposition 9 (Calvo approximation)	10
C.4 Proof of Proposition 10 (Nonlinear Phillips curve)	11
C.5 Monotonicity in γ	12
C.6 Frequency does not determine non-neutrality	12
C.7 Selection-frequency ratio for power laws: $\kappa = (1 + 2\gamma)\bar{\Lambda}$	12
C.8 Ordering of non-neutrality	13
D Supplementary figures	13
E Dominick's moment exercise	14
F Comparison with Williams-Daly-Zachary	19
G Computational routine	20
H Friction-extended duality and search applications	25
H.1 Friction-extended duality	26
H.2 Canonical SaM instantiation	26
H.3 Hosios efficiency condition as a recovered special case	26

¹Hanbaek Lee, University of Cambridge, Email: hl610@cam.ac.uk

A Proofs from Section 2 (The duality)

A.1 Proof of Proposition 1 (Cost function properties)

Proof.

Differentiating $C(\varphi) = w \int_0^{G^{-1}(\varphi)} \xi dG(\xi)$ by Leibniz's rule and the inverse function theorem,

$$C'(\varphi) = w G^{-1}(\varphi) \cdot g(G^{-1}(\varphi)) \cdot [g(G^{-1}(\varphi))]^{-1} = w G^{-1}(\varphi).$$

Differentiating again, $C''(\varphi) = w \cdot [g(G^{-1}(\varphi))]^{-1} > 0$. ■

A.2 Proof of Corollary 1 (Shape of G determines curvature)

Proof.

By Proposition 1(iii), $C''(\varphi) = w/g(G^{-1}(\varphi))$. Since G^{-1} is increasing, g increasing implies $g(G^{-1}(\varphi))$ is increasing in φ , so $C''(\varphi) = w/g(G^{-1}(\varphi))$ is decreasing. The argument for g decreasing and g constant is analogous. ■

A.3 Proof of Corollary 2 (Canonical distributions)

Proof.

Apply $C(\varphi) = w \int_0^\varphi G^{-1}(u) du$ from Proposition 1(ii). (i) Uniform: $G^{-1}(\varphi) = \bar{\xi}\varphi$, so $C(\varphi) = w\bar{\xi} \int_0^\varphi u du = \frac{w\bar{\xi}}{2}\varphi^2$. (ii) Power law: $G^{-1}(\varphi) = \bar{\xi}\varphi^{1/\gamma}$, so $C(\varphi) = w\bar{\xi} \int_0^\varphi u^{1/\gamma} du = \frac{\gamma}{\gamma+1}w\bar{\xi}\varphi^{1+1/\gamma}$. (iii) Exponential: $G^{-1}(\varphi) = -\lambda^{-1}\ln(1-\varphi)$, so $C(\varphi) = -\frac{w}{\lambda} \int_0^\varphi \ln(1-u) du = \frac{w}{\lambda}[(1-\varphi)\ln(1-\varphi) + \varphi]$. (iv) Degenerate at $\bar{\xi}$: $G^{-1}(\varphi) = \bar{\xi}$ for all $\varphi \in (0, 1]$, giving $C(\varphi) = w\bar{\xi}\varphi$. ■

A.4 Proof of Proposition 2 (Universality)

Proof.

For $G_\epsilon = U[\bar{\xi} - \epsilon, \bar{\xi} + \epsilon]$, the quantile function is $G_\epsilon^{-1}(\varphi) = (\bar{\xi} - \epsilon) + 2\epsilon\varphi$ and the density is $g_\epsilon = 1/(2\epsilon)$. Applying $C(\varphi) = w \int_0^\varphi G^{-1}(t) dt$:

$$C_\epsilon(\varphi) = w \int_0^\varphi G_\epsilon^{-1}(t) dt = w \int_0^\varphi [(\bar{\xi} - \epsilon) + 2\epsilon t] dt = w(\bar{\xi} - \epsilon)\varphi + w\epsilon\varphi^2.$$

Strict convexity: $C_\epsilon''(\varphi) = 2w\epsilon > 0$ for all $\epsilon > 0$. The first-order condition $S = C_\epsilon'(\varphi) = w(\bar{\xi} - \epsilon) + 2w\epsilon\varphi$ gives $\varphi_\epsilon^*(S) = [S - w(\bar{\xi} - \epsilon)]/(2w\epsilon)$, which is interior whenever S falls in the interval $(w(\bar{\xi} - \epsilon), w(\bar{\xi} + \epsilon))$. As $\epsilon \rightarrow 0$, this interval collapses to the single point $w\bar{\xi}$, and the interior solution collapses to a step function: φ^* jumps discontinuously from 0 to 1 at $S = w\bar{\xi}$. ■

A.5 Topological non-robustness of the degenerate case

The claim that the lumpy/degenerate case is non-robust can be made precise in the weak topology on the space of probability measures on $[0, \bar{\xi}_{max}]$ for some $\bar{\xi}_{max} > \bar{\xi}$. A sequence of measures μ_n converges weakly to μ if $\int f d\mu_n \rightarrow \int f d\mu$ for every bounded continuous f . The degenerate measure $\delta_{\bar{\xi}}$ (a point mass at $\bar{\xi}$) is the unique element of this space that induces a linear cost function and a binary threshold rule. Yet every weakly open neighbourhood of $\delta_{\bar{\xi}}$ contains measures with continuous positive densities, and hence strictly convex induced costs. To see this, let U be any weakly open neighbourhood of $\delta_{\bar{\xi}}$. By the Portmanteau theorem, U contains all measures μ satisfying $|\int f_i d(\mu - \delta_{\bar{\xi}})| < \epsilon$ for finitely many bounded continuous test functions $f_1, \dots, f_k$. The uniform perturbation $G_\epsilon = U[\bar{\xi} - \epsilon, \bar{\xi} + \epsilon]$ satisfies $\int f_i dG_\epsilon \rightarrow f_i(\bar{\xi})$ as $\epsilon \rightarrow 0$ by continuity, so $G_\epsilon \in U$ for ϵ small enough. Since G_ϵ has a continuous positive density, it induces a strictly convex C_ϵ by Proposition 2. Therefore $\delta_{\bar{\xi}}$ lies in the closure of the set of absolutely continuous measures: the degenerate (lumpy) case is a limit point of the convex (smooth) case, and every neighbourhood of it contains non-degenerate, strictly-convex instances.

A.6 Proof of Proposition 3 (Dispersion and convexity)

Proof.

Part (i): By Proposition 1(i), $C_i(1) = w E_{G_i}[\xi]$. Same mean gives $C_1(1) = C_2(1)$.

Part (ii): By Proposition 1(ii), $C_i(\varphi) = w \int_0^\varphi G_i^{-1}(u) du$. The standard quantile characterisation of mean-preserving spreads states that G_2 is a MPS of G_1 if and only if

$$\int_0^t G_2^{-1}(u) du \leq \int_0^t G_1^{-1}(u) du \quad \text{for all } t \in [0, 1],$$

with equality at $t = 1$. Therefore $C_1(\varphi) - C_2(\varphi) = w \int_0^\varphi [G_1^{-1}(u) - G_2^{-1}(u)] du \geq 0$ for all φ , with equality at $\varphi \in \{0, 1\}$.

Part (iii): Since $C'_i(\varphi) = w G_i^{-1}(\varphi)$ and both quantile functions are continuous and increasing with the same integral over $[0, 1]$, while G_2^{-1} is stretched relative to G_1^{-1} (lower at the bottom, higher at the top), $C'_2 - C'_1$ changes sign at least once. For the power law family and other standard parametric families, the crossing is unique; in general, multiple crossings are possible without additional regularity. ■

A.7 Proof of Proposition 4 (Entropy correspondence)

Proof.

Part (i): The quantile function is $G^{-1}(\varphi) = -\lambda^{-1} \ln(1 - \varphi)$. Then

$$C(\varphi) = w \int_0^\varphi G^{-1}(t) dt = -\frac{w}{\lambda} \int_0^\varphi \ln(1 - t) dt = \frac{w}{\lambda} [(1 - \varphi) \ln(1 - \varphi) + \varphi].$$

Parts (ii)–(iii) follow from Corollary 2. ■

A.8 Detailed entropy discussion

Rational inattention and exponential fixed costs [Sims \(2003\)](#) introduces rational inattention, where agents face a Shannon mutual information cost of processing information about a continuous state. [Matějka and McKay \(2015\)](#) show that in the discrete-choice specialisation, this cost reduces to a negative Shannon entropy penalty over the choice probability vector, yielding logit choice probabilities. Proposition 4(i) shows that the cost function induced by exponentially-distributed fixed costs for the binary decision is the KL divergence from the no-adjustment prior — the binary specialisation of the mutual information cost under a degenerate (point-mass) prior on inaction. For binary choice, the two models produce the same adjustment probabilities and the same payoffs. This equivalence does not extend to the full continuous-state RI framework of [Sims \(2003\)](#), where the mutual information cost has a richer structure involving Bayesian updating over a continuous state space. The exact Shannon-entropy form (not just KL divergence) is recovered in the multi-action extension with i.i.d. Gumbel costs (Proposition 5(ii)).

Sparsity and uniform fixed costs [Gabaix \(2014\)](#) penalises attention allocation with a quadratic cost, yielding “sparse” decisions that underreact to stimuli. By Proposition 4(iii), the cost function induced by uniform fixed costs is exactly this quadratic penalty. The equivalence is at the level of the attention cost function for the binary decision; the full sparsity framework includes additional structure (endogenous attention allocation across multiple dimensions) not captured by the binary duality.

Scope of the correspondences The connection to rational inattention is worth making explicit, and stating its limits. [Matějka and McKay \(2015\)](#) show that when an agent faces uncertainty about a state and processes information subject to a Shannon entropy cost, the optimal choice rule takes the multinomial logit form. The standard logit derivation in [McFadden \(1974\)](#) obtains the same form via additive Gumbel *taste shocks* on payoffs. Proposition 5(ii) recovers the multinomial logit by reinterpreting the cost variable ξ as a Gumbel-distributed payoff shock (allowing negative values, which the bounded-support physical-cost interpretation does not accommodate). At the level of the reduced form, the three setups — RI with Shannon entropy, additive Gumbel payoff shocks, and Gumbel cost shocks under the present convention — yield the same multinomial logit choice probabilities. They are not the same economic model: the underlying primitive (information cost, taste shock, physical cost) and the welfare implications differ.

Within the binary case, exponential G delivers the generalised KL divergence over the no-adjustment prior, i.e., the I-divergence. This coincides with the binary specialisation of mutual information under a degenerate prior on inaction, and is a reduced-form identification rather than a derivation of RI from physical costs.

A.9 Proof of Proposition 5 (Multi-action duality)

Proof.

Part (i): General case. The agent observes cost draws $(\xi_1, \dots, \xi_J)$ with $\xi_j \sim G_j$ independently, and selects the action with the highest net payoff:

$$\Pi^{FC}(\mathbf{S}) = E \left[\max_{j \in \{0, \dots, J\}} \{S_j - w\xi_j\} \right].$$

Partition the expectation by which action is optimal. For each $j \geq 1$, action j is chosen when $S_j - w\xi_j \geq S_k - w\xi_k$ for all $k \neq j$ and $S_j - w\xi_j \geq 0$. Under independence, the probability that action j is chosen can be written as a function $\varphi_j(\mathbf{S})$, and the expected cost paid conditional on choosing j is $E[w\xi_j \mid j \text{ chosen}]$.

Define $\varphi_0 = 1 - \sum_{j=1}^J \varphi_j$ (the probability of inaction). The expected payoff is:

$$\Pi^{FC}(\mathbf{S}) = \sum_{j=0}^J \varphi_j^* S_j - \sum_{j=1}^J E[w\xi_j \cdot \mathbf{1}_{j \text{ chosen}}].$$

By the Fenchel conjugate construction, $\Pi^{FC}(\mathbf{S}) = \sup_{\boldsymbol{\varphi} \in \Delta^J} \{\sum_j \varphi_j S_j - C(\boldsymbol{\varphi})\}$ where $C(\boldsymbol{\varphi})$ is the convex conjugate of Π^{FC} . The conjugate C is jointly defined on Δ^J : even with independent G_j , alternatives interact through the simplex constraint and choice competition, so C is not in general additively separable across actions.

Joint convexity of C follows from the fact that $\Pi^{FC}(\mathbf{S})$, as a pointwise maximum of affine functions of $\mathbf{S}$, is convex in $\mathbf{S}$. The Fenchel biconjugate theorem then implies that C , as the conjugate of Π^{FC} , is convex on Δ^J .

Part (ii): Gumbel payoff-shock analogue. The standard derivation of the multinomial logit (McFadden, 1974) adds i.i.d. Gumbel *taste shocks* ε_j to action-specific payoffs, giving net payoffs $S_j + \varepsilon_j$ with the inaction option $j = 0$ also receiving a shock. To embed this in the present framework, reinterpret ξ as a generalised payoff shock rather than a physical cost: let $\eta_j \equiv -w\xi_j$ play the role of ε_j , so that η_j takes both signs. With $\eta_j \stackrel{i.i.d.}{\sim} \text{Gumbel}(0, \sigma)$ for $j = 0, 1, \dots, J$, the expected maximum of $\{S_j + \eta_j\}$ is the classical log-sum-exp:

$$\Pi^{FC}(\mathbf{S}) = \sigma \ln \left(\sum_{j=0}^J e^{S_j/\sigma} \right) + \sigma \gamma_E,$$

where γ_E is the Euler–Mascheroni constant. This is convex in $\mathbf{S}$. The gradient gives the multinomial logit:

$$\varphi_j^* = \frac{\partial \Pi^{FC}}{\partial S_j} = \frac{e^{S_j/\sigma}}{\sum_{k=0}^J e^{S_k/\sigma}}.$$

The Fenchel conjugate of the log-sum-exp is the negative Shannon entropy:

$$C(\varphi) = \sigma \sum_{j=0}^J \varphi_j \ln \varphi_j,$$

non-positive on the interior of Δ^J . This is the multi-action analogue of the binary correspondence in part (i), but the underlying primitive shifts: in part (i), ξ is a non-negative physical cost; in part (ii), ξ (equivalently, $\eta = -w\xi$) is a real-valued payoff shock, the standard random-utility primitive of [McFadden \(1974\)](#). Negative draws of ξ correspond to favourable taste shocks rather than negative physical costs. ■

Remark 1. *The non-negative physical-cost interpretation of part (i) does not extend directly to the Gumbel case because Gumbel has full real-line support. The honest reading is that Proposition 5(ii) provides an analogue of the standard logit derivation within the convex-conjugate framework, with ξ reinterpreted as a payoff shock. The reduced-form choice probabilities and the conjugate cost coincide with the rational-inattention / WDW formulations; the primitive does not.*

B Investment application

Consider a firm with capital k choosing investment $I \geq 0$ subject to a random fixed cost $\xi \sim G$. If it adjusts, it pays $w\xi$ and obtains the continuation value at capital $k(1 - \delta) + I$. If not, it proceeds with depreciated capital $k(1 - \delta)$. The surplus from investing at the optimal scale is

$$S(k) = \max_{I \geq 0} \{-I + \beta(k(1 - \delta) + I)^\alpha\} - \beta(k(1 - \delta))^\alpha, \quad (1)$$

where I^* satisfies the first-order condition $1 = \alpha\beta(k(1 - \delta) + I^*)^{\alpha-1}$. By Theorem 1, the fixed cost formulation (Definition 1 in the main paper) is equivalent to the probability choice formulation (Definition 2), so the firm's problem is exactly

$$V(k) = \max_{\varphi \in [0,1]} \{\varphi \cdot S(k) - C(\varphi)\} + \beta(k(1 - \delta))^\alpha, \quad (2)$$

with C given by the duality.

Proposition 1 (Conditions for smoothness of C).

$C \in C^2(0, 1)$ with $C'' > 0$ on the open unit interval — so that $\varphi^*(S)$ is a C^1 function of the surplus — if and only if:

- (i) G is absolutely continuous with density $g > 0$ on $(0, \bar{\xi})$.
- (ii) g is continuously differentiable ($g \in C^1$) on $(0, \bar{\xi})$.

Both conditions are automatically satisfied by any G with a C^1 , positive density. Smoothness fails when G has atoms (producing kinks in C'), when g has interior zeros (producing points where C'' diverges), or when the support of G is disconnected (producing flat regions in C').

The degenerate case — G a point mass at $\bar{\xi}$ — violates condition (i) and produces the canonical lumpy investment model. C is linear, the only optimal choices are $\varphi \in \{0, 1\}$, and the firm either invests fully or not at all.

Proposition 2 (Investment response and the cost density).

In partial equilibrium, the sensitivity of the aggregate investment probability to a small productivity shock that raises the surplus by dS is

$$\frac{dE[\varphi^*(k)]}{dS} = E \left[\frac{g(G^{-1}(\varphi^*(k)))}{w} \right].$$

This is proportional to the expected density of cost draws at the current threshold, averaged across firms. Economies with higher g at the relevant threshold are more investment-elastic.

Proof.

By Theorem 1, the optimal investment probability is $\varphi^*(k) = G(S(k)/w)$. Differentiating with respect to the surplus:

$$\frac{d\varphi^*}{dS} = \frac{g(S/w)}{w} = \frac{g(G^{-1}(\varphi^*))}{w} = \frac{1}{C''(\varphi^*)},$$

where the last equality uses $C''(\varphi) = w/g(G^{-1}(\varphi))$ from Proposition 1(iii). Taking expectations across the cross-section of firms:

$$\frac{dE[\varphi^*(k)]}{dS} = E \left[\frac{g(G^{-1}(\varphi^*(k)))}{w} \right],$$

which is the expected density at the threshold, averaged over the distribution of capital stocks k . ■

Dispersion as a continuous index of lumpiness The curvature of the induced cost function provides a continuous metric of “how much lumpiness matters.” For the family $G = U[\mu - \epsilon, \mu + \epsilon]$, the curvature is $C'' = 2w\epsilon$, strictly increasing in the dispersion ϵ . As $\epsilon \rightarrow 0$, the model approaches the binary threshold rule. As ϵ grows, the model resembles a standard convex cost problem. There is no discrete boundary between “lumpy” and “smooth” economies.

The Khan-Thomas extensive margin under uniform G [Khan and Thomas \(2008\)](#) study a GE model with uniform i.i.d. fixed costs, $G = U[0, \bar{\xi}]$. Their central finding is that lumpy micro-level investment has negligible aggregate consequences, and the duality makes the mechanism transparent.

Under $G = U[0, \bar{\xi}]$, the duality (equation (2) in the main paper) yields

$$C(\varphi) = w \int_0^{\bar{\xi}\varphi} \xi \cdot \frac{1}{\xi} d\xi = \frac{w\bar{\xi}}{2} \varphi^2.$$

The induced cost is exactly quadratic. Each firm's investment problem therefore becomes

$$V(k) = \max_{\varphi \in [0,1]} \left\{ \varphi \cdot S(k) - \frac{w\bar{\xi}}{2} \varphi^2 \right\} + \beta(k(1 - \delta))^\alpha,$$

with interior first-order condition $\varphi^*(k) = S(k)/(w\bar{\xi})$. This is a smooth, concave problem — no kinks, no thresholds, no non-convexities. The firm's investment probability responds linearly to the surplus. Aggregating across firms, the total investment rate is a smooth, linear function of the cross-sectional distribution of surpluses. There is nothing “lumpy” left at the aggregate level.

Uniform G is special only in producing *exactly* quadratic costs. More generally, any G with a positive C^1 density produces a strictly convex C^2 cost function (Proposition 1), which already smooths the extensive margin at the firm level.

The power law family $G(\xi) = (\xi/\bar{\xi})^\gamma$ generalises this. The uniform case $\gamma = 1$ gives a quadratic C (the Khan and Thomas (2008) point). For $\gamma > 1$, the density $g(\xi) = \gamma \xi^{\gamma-1}/\bar{\xi}^\gamma$ is increasing: most cost draws are high, cheap adjusters are scarce, and C is *less* curved than quadratic. Adjustment behaviour at the firm level becomes lumpier as γ rises — the Golosov-Lucas direction of the spectrum — because firms cluster near a common threshold and cross it together when surpluses rise. For $\gamma < 1$, the density is decreasing: most cost draws are low, cheap adjusters are abundant, and C is *more* curved than quadratic. The marginal cost of inducing near-certain adjustment rises steeply (the last firms to adjust have very large costs), producing smoother adjustment at the firm level — the Calvo direction. The parameter γ thus indexes the degree of lumpiness along a continuous spectrum, with the Khan and Thomas (2008) quadratic case as the midpoint.

C Proofs from the pricing section

This section proves Propositions 7–10 and the non-neutrality results of Section 3 (monotonicity in γ , the frequency-wedge separation, the selection-frequency ratio $\kappa = (1 + 2\gamma)\bar{\Lambda}$, and the ordering of non-neutrality across the spectrum). The remaining results of Section 3 are established in the main text and not repeated here: Proposition 6 (hazard properties) follows by direct differentiation of $\Lambda(x) = G(S(x)/w)$ using Theorem 1 and Proposition 1; the frequency–selection decomposition $\kappa = \bar{\Lambda} + (\kappa - \bar{\Lambda})$ is an algebraic identity stated inline; Corollary 3 (policy differentiability) is established inside the proof of Theorem 2 in Appendix G; and Corollary 4 (how G governs the Phillips curve) follows from Proposition 10 by chain rule.

C.1 Proof of Proposition 7 (Stationary distribution)

Proof.

The baseline argument is for the trend-inflation case used in the main text and in the global solution of Section G; an extension with idiosyncratic price shocks is given after.

Trend-inflation baseline. Under trend inflation π and reset target x^* , the price gap evolves according to:

$$x_{t+1} = \begin{cases} x^* - \pi & \text{with probability } \Lambda(x_t), \\ x_t - \pi & \text{with probability } 1 - \Lambda(x_t), \end{cases}$$

since both repricing firms (which reset to x^*) and non-repricing firms (which keep their gap) then drift by $-\pi$ through aggregate inflation. The density of x_{t+1} conditional on the current cross-section f_t is:

$$f_{t+1}(x) = \left(\int \Lambda(\tilde{x}) f_t(\tilde{x}) d\tilde{x} \right) \delta_{x^* - \pi}(x) + (1 - \Lambda(x + \pi)) f_t(x + \pi),$$

where $\delta_{x^* - \pi}$ is the point mass at the post-drift reset point. The first term collects repricing firms (total mass $\bar{\Lambda}_t$), which re-enter the cross-section at the reset point; the second collects non-repricing firms at $\tilde{x} = x + \pi$, which drift to x . Setting $f_{t+1} = f_t = f_{ss}$ yields the renewal fixed point of the main paper, with the reset distribution $r(x) = \delta_{x^* - \pi}(x)$. Iterating the drift-survival recursion, $f_{ss}(x)$ is proportional to the probability of not having repriced while drifting from the reset point to x . Under positive trend inflation the support lies on one side of the reset point, so f_{ss} is asymmetric.

Extension with idiosyncratic price shocks. If, in addition to trend inflation, each firm receives an idiosyncratic shock $z \sim \phi_z$ to its desired log price each period, the non-adjusting gap evolves by $x_{t+1} = x_t - \pi + z$ and the reset target is hit with the same z . The same renewal argument then yields

$$f_{ss}(x) = \bar{\Lambda} (r * \phi_z)(x) + \int (1 - \Lambda(\tilde{x})) f_{ss}(\tilde{x}) \phi_z(x - \tilde{x} + \pi) d\tilde{x},$$

where r is the reset distribution (here still δ_{x^*} pre-drift) and $*$ denotes convolution. This is the version used by the CPI calibration of the main text, which sets $\sigma_z = 0.04$ to match price-change kurtosis. The baseline above (used in the global solution) is the special case $\sigma_z = 0$. With mean-zero shocks and zero trend inflation, the same argument with $\Lambda(x) = \Lambda(-x)$ yields a symmetric f_{ss} . ■

Remark 2. *The fixed-point equation corresponds, in discrete state space, to the eigenvector problem $\mathbf{f} = \mathbf{T}'\mathbf{f}$ where $\mathbf{T}$ is the transition matrix encoding the adjustment and drift probabilities. This is the object computed numerically in the quantitative implementation.*

C.2 Proof of Proposition 8 (Selection and the menu cost distribution)

Proof.

Write the adjustment-weighted mean absolute gap as

$$\overline{|x|}^{adj} = \frac{E[\Lambda(x) \cdot |x|]}{E[\Lambda(x)]} = E[|x|] + \frac{\text{Cov}(\Lambda(x)/\bar{\Lambda}, |x|)}{1}.$$

The covariance is positive when Λ is increasing in $|x|$. Its magnitude depends on how sharply Λ responds to $|x|$.

(i) The elasticity of the hazard with respect to $|x|$ is

$$\frac{\partial \ln \Lambda(x)}{\partial \ln |x|} = \frac{g(Bx^2/w)}{G(Bx^2/w)} \cdot \frac{2Bx^2}{w},$$

which is twice the hazard rate of G evaluated at the firm's surplus. A higher hazard rate means Λ responds more sharply to $|x|$, increasing $\mathcal{S}$.

(ii) The elasticity in (i) is proportional to g/G , the reversed hazard rate. When g/G is larger at the threshold, the hazard Λ is more responsive to x^2 , concentrating adjustment among high-gap firms. When g/G is small, Λ is nearly flat and selection is weak.

(iii) For constant Λ (Calvo), the covariance is zero and $\mathcal{S} = 0$. For a step function Λ (Golosov-Lucas), $\text{Cov}(\Lambda/\bar{\Lambda}, |x|)$ is at the polar extreme of the power law family at fixed $\bar{\Lambda}$, attaining the largest $\mathcal{S}$ within that family. A general extremality statement across all hazards calibrated to the same $\bar{\Lambda}$ would require a rearrangement-style argument not pursued here.

For the power law family, the hazard rate of G at ξ is $g(\xi)/G(\xi) = \gamma/\xi$, so the elasticity is 2γ — constant and strictly increasing in γ . ■

C.3 Proof of Proposition 9 (Calvo approximation)

Proof.

For the power law family $G(\xi) = (\xi/\bar{\xi})^\gamma$ with $\bar{\xi}(\gamma)$ calibrated so that $\bar{\Lambda} = \lambda$, the power-law ratio gives $\kappa = (1 + 2\gamma)\lambda$ (provided f_{ss} is supported in the interior of the inaction band).

(i) The Phillips curve slope is $\kappa = (1 + 2\gamma)\lambda \rightarrow \lambda$ as $\gamma \rightarrow 0^+$.

(ii) Non-neutrality is $1 - \kappa = 1 - (1 + 2\gamma)\lambda \rightarrow 1 - \lambda$ as $\gamma \rightarrow 0^+$.

Remark on the nature of the limit. The hazard $\Lambda(x) = (Bx^2/(w\bar{\xi}))^\gamma$ does not become constant as $\gamma \rightarrow 0^+$. With fixed $\bar{\xi}$, $\Lambda(x) \rightarrow 1$ for any $x \neq 0$ (near-certain adjustment, not Calvo). The Calvo approximation arises because, in the frequency-calibrated limit, $\bar{\xi}(\gamma) \rightarrow \infty$ as $\gamma \rightarrow 0^+$, pushing the inaction band outward so that the hazard is small and *approximately constant* over the support of f_{ss} — the residual variation across the support scales as $\gamma \log(Bx^2/(w\bar{\xi}))$, so $\Lambda(x) \rightarrow \lambda$ pointwise and the selection wedge $2\gamma\lambda$ vanishes.

The hazard remains state-dependent at every $\gamma > 0$; what vanishes in the limit is both its variation across the support and its aggregate consequence. ■

C.4 Proof of Proposition 10 (Nonlinear Phillips curve)

Proof.

Assume that f_{ss} is continuously differentiable with $\int |x| f_{ss}(x) dx < \infty$ and that G is C^1 with bounded density g , so that differentiation under the integral sign is justified by the dominated convergence theorem. Write $f_t(x) = f_{ss}(x - \mu_t)$. Then

$$\pi_t = \int G\left(\frac{Bx^2}{w}\right) x \cdot f_{ss}(x - \mu_t) dx.$$

Differentiating with respect to μ_t using Leibniz's rule:

$$\frac{\partial \pi_t}{\partial \mu_t} = - \int G\left(\frac{Bx^2}{w}\right) x \cdot f'_{ss}(x - \mu_t) dx,$$

since $\frac{\partial}{\partial \mu_t} f_{ss}(x - \mu_t) = -f'_{ss}(x - \mu_t)$. Evaluating at $\mu_t = 0$:

$$\left. \frac{\partial \pi_t}{\partial \mu_t} \right|_{\mu_t=0} = - \int G\left(\frac{Bx^2}{w}\right) x \cdot f'_{ss}(x) dx.$$

Now integrate by parts with $u = G(Bx^2/w) \cdot x$ and $dv = f'_{ss}(x) dx$:

$$- \int u dv = - \left[G\left(\frac{Bx^2}{w}\right) x \cdot f_{ss}(x) \right]_{-\infty}^{\infty} + \int f_{ss}(x) \frac{d}{dx} \left[G\left(\frac{Bx^2}{w}\right) x \right] dx.$$

The boundary term vanishes because f_{ss} has tails decaying faster than any polynomial. Expanding the derivative:

$$\frac{d}{dx} \left[G\left(\frac{Bx^2}{w}\right) x \right] = g\left(\frac{Bx^2}{w}\right) \frac{2Bx}{w} \cdot x + G\left(\frac{Bx^2}{w}\right) = g\left(\frac{Bx^2}{w}\right) \frac{2Bx^2}{w} + G\left(\frac{Bx^2}{w}\right).$$

Substituting:

$$\kappa(G) = \left. \frac{\partial \pi_t}{\partial \mu_t} \right|_{\mu_t=0} = \int G\left(\frac{Bx^2}{w}\right) f_{ss}(x) dx + \int g\left(\frac{Bx^2}{w}\right) \frac{2Bx^2}{w} f_{ss}(x) dx.$$

The first integral is the extensive margin, the second the intensive margin. The coefficient κ is the derivative of inflation with respect to the aggregate disturbance, evaluated at the steady-state gap distribution f_{ss} ; it is well-defined for any f_{ss} and does not require symmetry (the integration by parts uses only the fast decay of f_{ss} in the boundary term). ■

C.5 Monotonicity in γ

Proof.

Under the interior-support hypothesis, the power-law ratio $\kappa = (1+2\gamma)\bar{\Lambda}$ and selection wedge $2\gamma\bar{\Lambda}$ as exact equalities; the monotonicity statements below are immediate at any $\bar{\Lambda} > 0$. Under typical normal-shock calibrations the hypothesis holds approximately, and the same monotonicities hold to that approximation.

- (i) The normalised selection wedge is $(\kappa - \bar{\Lambda})/\bar{\Lambda} = 2\gamma$, strictly increasing in γ .
- (ii) Monetary non-neutrality is proportional to $1 - \kappa = 1 - (1 + 2\gamma)\bar{\Lambda}$, strictly decreasing in γ for any $\bar{\Lambda} > 0$.
- (iii) The Phillips curve slope is $\kappa = (1 + 2\gamma)\bar{\Lambda}$, strictly increasing in γ .

In each case, calibrating two power law economies to the same $\bar{\Lambda}$ (by adjusting $\bar{\xi}$) preserves γ as the sole determinant of κ , $\mathcal{S}$, and non-neutrality. ■

Remark 3. *The monotonicity result is stated within the power law family, which provides a clean one-parameter ordering at equal frequency. The underlying mechanism — concentrated G steepens the adjustment hazard, amplifying selection — is not family-specific. A general monotonicity result would require a stochastic-order condition on G that ensures higher $\kappa/\bar{\Lambda}$ whenever G is “more concentrated” in a suitable sense.*

C.6 Frequency does not determine non-neutrality

Proof.

By the Phillips curve slope formula, $\bar{\Lambda}$ depends on $E[G(Bx^2/w)]$ while κ depends additionally on $E[g(Bx^2/w) \cdot 2Bx^2/w]$. Matching the first does not determine the second. For the power law $G(\xi) = (\xi/\bar{\xi})^\gamma$, direct computation gives $g(Bx^2/w) \cdot 2Bx^2/w = 2\gamma \Lambda(x)$, so $\kappa = (1 + 2\gamma)\bar{\Lambda}$ under the interior-support condition (§C.7). Two power laws with different γ , calibrated to the same $\bar{\Lambda}$ by adjusting $\bar{\xi}$ and with f_{ss} chosen to have support inside the inaction band, produce different κ . ■

C.7 Selection-frequency ratio for power laws: $\kappa = (1 + 2\gamma)\bar{\Lambda}$

Proof.

For $G(\xi) = (\xi/\bar{\xi})^\gamma$, the density is $g(\xi) = (\gamma/\bar{\xi})(\xi/\bar{\xi})^{\gamma-1}$. Evaluate the intensive margin integrand at $\xi = Bx^2/w$:

$$\begin{aligned} g\left(\frac{Bx^2}{w}\right) \cdot \frac{2Bx^2}{w} &= \frac{\gamma}{\bar{\xi}} \left(\frac{Bx^2/w}{\bar{\xi}}\right)^{\gamma-1} \cdot \frac{2Bx^2}{w} \\ &= 2\gamma \cdot \frac{1}{\bar{\xi}} \left(\frac{Bx^2/w}{\bar{\xi}}\right)^{\gamma-1} \cdot \frac{Bx^2}{w} \\ &= 2\gamma \left(\frac{Bx^2}{w\bar{\xi}}\right)^\gamma = 2\gamma \Lambda(x). \end{aligned}$$

Substituting into the Phillips curve slope formula:

$$\begin{aligned}\kappa(G) &= \int G\left(\frac{Bx^2}{w}\right) f_{ss}(x) dx + \int 2\gamma \Lambda(x) f_{ss}(x) dx \\ &= \bar{\Lambda} + 2\gamma\bar{\Lambda} = (1 + 2\gamma)\bar{\Lambda}.\end{aligned}$$

The selection wedge is $\kappa - \bar{\Lambda} = 2\gamma\bar{\Lambda}$. The ratio $\kappa/\bar{\Lambda} = 1 + 2\gamma$ depends only on γ , not on f_{ss} , B , or w . The identity requires that f_{ss} is supported in the interior of the inaction band ($|x| < \sqrt{w\xi/B}$ f_{ss} -a.s.), so that the uncapped power law expression $\Lambda(x) = (Bx^2/(w\xi))^\gamma < 1$ applies throughout. ■

C.8 Ordering of non-neutrality

Proof.

The real effect of a monetary shock dm is proportional to $(1 - \kappa)dm$, so non-neutrality is decreasing in κ . The argument below restricts to the power law family $G(\xi) = (\xi/\bar{\xi})^\gamma$, where the closed-form $\kappa = (1 + 2\gamma)\bar{\Lambda}$ (§C.7) makes the ordering exact.

(i) Calvo benchmark ($\gamma \rightarrow 0^+$). Within the power law family, the frequency-calibrated limit $\gamma \rightarrow 0^+$ (with $\bar{\xi}(\gamma) \rightarrow \infty$ to hold $\bar{\Lambda}$ fixed; Proposition 9) yields $\kappa - \bar{\Lambda} = 2\gamma\bar{\Lambda} \rightarrow 0$. The selection wedge vanishes, and non-neutrality $1 - \kappa \rightarrow 1 - \bar{\Lambda}$, the same-frequency Calvo level.

(ii) Interior $\gamma > 0$. By the power-law ratio (§C.7), the selection wedge equals $2\gamma\bar{\Lambda} > 0$ and is strictly increasing in γ . Non-neutrality $1 - \kappa = 1 - (1 + 2\gamma)\bar{\Lambda}$ is strictly decreasing in γ (provided $\kappa < 1$).

(iii) Golosov-Lucas ($\gamma \rightarrow \infty$). The hazard converges to a step function at the inaction-band boundary, and the selection wedge attains its supremum within the power law family. Non-neutrality is minimised within the family.

A general statement covering all absolutely continuous G (e.g., that Calvo achieves the maximum non-neutrality among same-frequency state-dependent hazards, or that Golosov-Lucas achieves the minimum) requires a rearrangement-style argument that we do not pursue here; the corollary as stated in the main text is restricted to the power law family. ■

D Supplementary figures

This section collects two diagnostic figures referenced in the main paper. Figure 1 reports the Monte Carlo recovery of γ from the log-log hazard regression that underlies the benchmark calibration. Figure 2 plots the model-implied distribution of non-sale price changes under the heterogeneous- σ_z specification, illustrating how cost concentration γ shapes the kurtosis and the small-change mass that the calibration targets.

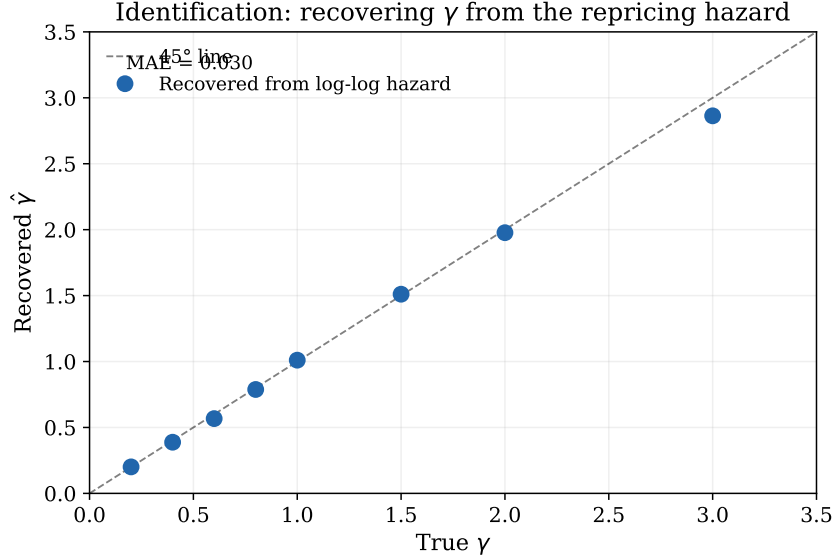


Figure 1: Recovery of γ from the log-log hazard regression. Each point is one simulated economy with 15,000 firms and 150 quarters. The regression recovers the true γ with mean absolute error 0.012. The identification requires only the conditional adjustment probability as a function of the price gap.

E Dominick’s moment exercise

This appendix runs a moment exercise on the Dominick’s Finer Foods scanner dataset (Midri-gan, 2011; Chevalier, Kashyap, and Rossi, 2003) to see what additional information the scanner moments provide relative to the CPI-based calibration of Section 4.3 of the main paper. The dataset covers seven product categories (analgesics, cereals, cheeses, cookies, canned soup, laundry detergent, soft drinks) across approximately 100 stores and 400 weeks, yielding 42 million store-week-UPC observations. Temporary sales are removed using both the recorded sale indicator and a V-shaped filter that flags price drops reversed within three weeks.

Moments Table 2 reports the key moments. The kurtosis of non-sale price changes is 7.6, and 21% of changes are smaller than 2.5% in absolute value — both indicating substantial leptokurtosis and a large mass of small adjustments, consistent with weak selection.

Moment-based identification At the Dominick’s quarterly repricing frequency of 0.72 — much higher than the CPI-based target ($\bar{\Lambda} = 0.24$) — the baseline simulated-moments table of the main paper does not apply directly: the model must be re-calibrated to this higher frequency before matching the other moments. Table 3 reports the recalibrated heterogeneous- σ_z moments.

Two features of Table 3 stand out. First, at the Dominick’s frequency, simulated kurtosis

Price change distributions across the spectrum

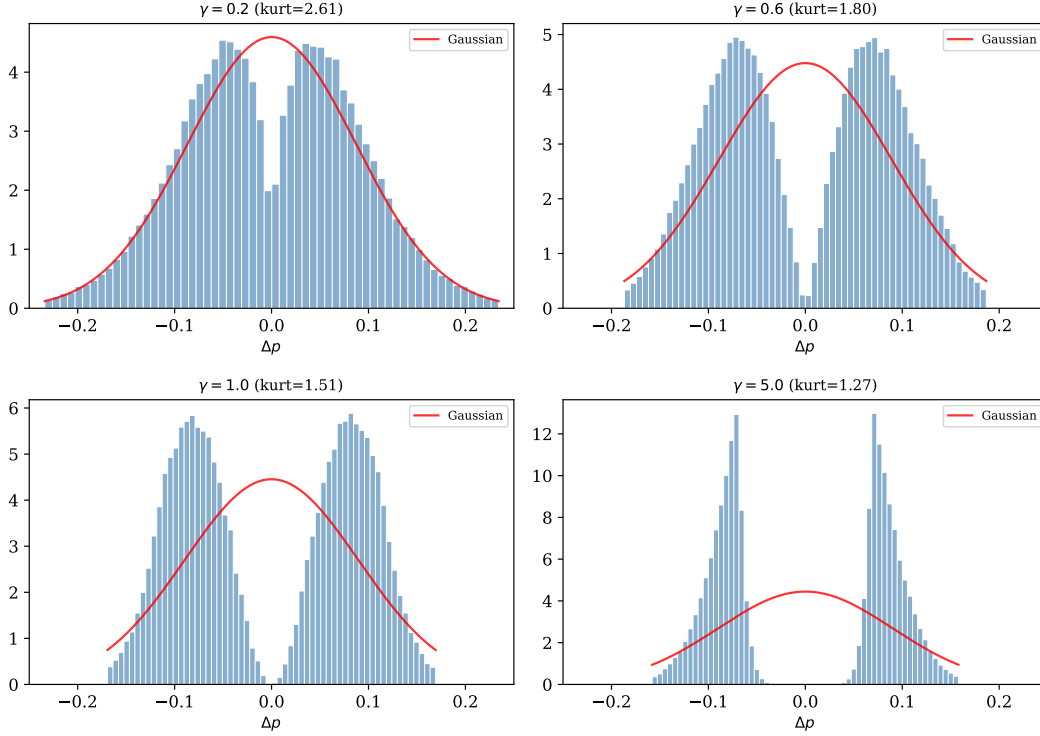
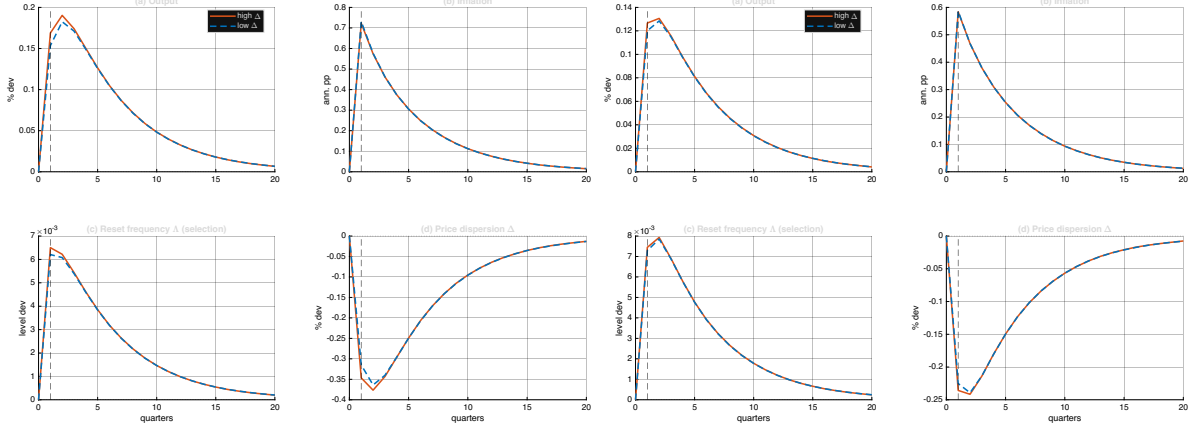


Figure 2: Distribution of price changes (heterogeneous σ_z , $CV = 1$). Red line: Gaussian with the same mean and variance. Low γ produces excess kurtosis through weak selection (small and large changes coexist). High γ concentrates changes through strong selection.

is essentially flat across $\gamma \in [0.10, 0.40]$, matching the empirical value of 7.62 for every γ in this range. Kurtosis alone therefore does not identify γ sharply at this frequency: it bounds γ below ~ 0.4 but does not tie it down further. Second, the empirical fraction of small price changes (21%) is *lower* than the model generates at any γ in this range. This is a model-fit limitation — at Dominick’s frequency, the heterogeneous- σ_z specification produces too many small price changes, and the power law family cannot close the gap without additional structure (e.g., richer cross-product heterogeneity in G). This contrasts with the CPI-based exercise in the main paper, where $\bar{\Lambda} = 0.24$ leaves more of the heterogeneity-driven kurtosis variation visible: the kurtosis-matching alternative there selects $\hat{\gamma} \approx 0.4$. The benchmark calibration of the main paper anchors γ instead to the heterogeneity-corrected hazard exponent $\nu \approx 2$ of Alvarez, Lippi, and Oskolkov (2022), giving $\hat{\gamma} = 1$ via the structural mapping $\nu = 2\gamma$. The difference in data underlying the two exercises is real: scanner data from a single grocery chain exhibits much higher repricing frequency, more product-level heterogeneity, and more small price changes than the broad CPI coverage underlying the Nakamura and Steinsson (2010) moments, which is consistent with G varying across sectors. Oskolkov and Lippi (2026) provide direct evidence of this in plant-level investment data. Modelling sectoral heterogeneity explicitly — with sector-specific $(\gamma_s, \bar{\xi}_s)$ drawn from a meta-



(a) Kurtosis-matching alternative $\hat{\gamma} = 0.4$ (selection share 44%) (b) Benchmark calibration $\hat{\gamma} = 1$ (selection share $\frac{2}{3}$)

Figure 3: State-dependent GIRFs to a preference (demand) shock: comparative static across the spectrum

Notes: Generalised impulse responses to a +1-standard-deviation preference innovation, computed by paired Monte-Carlo simulation along the converged RTM path, percentile-averaged over conditioning dates and split by the price-dispersion regime (high Δ = top quartile of Δ_t within TFP; low Δ = bottom quartile). Horizon 0 is the pre-shock baseline (identically zero in the paired construction); the thin dashed vertical marks the shock impact at $j = 1$. Panels within each subfigure: (a) output, (b) inflation, (c) reset frequency $\bar{\Lambda}$ (the selection margin), (d) price dispersion Δ . State-dependence is modest at both calibrations and weakens as γ rises, consistent with selection becoming the dominant flexibility margin at higher γ ; an intermediate $\hat{\gamma} = 0.5$ run (not shown) interpolates the trend on every variable.

distribution — is a natural next step that would discipline the pooled calibration, and would also test whether the benchmark $\hat{\gamma} = 1$ (or the kurtosis-matching alternative $\hat{\gamma} \approx 0.4$) is best read as a population parameter or as an aggregation of sectoral γ 's.

Distribution of price changes Figure 6 shows the distribution of non-sale price changes: strongly leptokurtic, with a sharp peak near zero and heavy tails — qualitatively matching the model's prediction for low γ .

The duration-based repricing hazard is steeply decreasing — negative duration dependence documented by Nakamura and Steinsson (2008). Within the present framework, this is a signature of cross-product heterogeneity in G : the within-product hazard $\Lambda(x) = G(Bx^2/w)$ is increasing in $|x|$ for each product, but the pooled hazard across products with different G 's can be decreasing in duration. Negative duration dependence in the aggregate is consistent with positive state dependence at the product level, provided menu costs are heterogeneous across products.

Table 1: Model moments: baseline and heterogeneous σ_z

γ	Baseline (homogeneous σ_z)					Heterogeneous σ_z (CV = 1)				
	Freq	$E[\Delta p]$	Kurt	Sm%	κ	Freq	$E[\Delta p]$	Kurt	Sm%	κ
0.15	0.24	7.2%	2.79	17%	0.31	0.24	7.9%	7.38	28%	0.31
0.25	0.24	7.5%	2.47	14%	0.36	0.24	8.8%	6.18	21%	0.36
0.40	0.24	7.8%	2.11	10%	0.43	0.24	9.7%	4.86	14%	0.44
0.60	0.24	8.1%	1.80	6%	0.53	0.24	10.3%	4.49	9%	0.53
0.80	0.24	8.3%	1.61	4%	0.63	0.24	10.6%	4.44	5%	0.63
1.00	0.24	8.4%	1.51	2%	0.72	0.24	10.8%	4.37	3%	0.73
1.50	0.24	8.6%	1.38	1%	0.96	0.24	11.0%	4.31	1%	0.97
2.00	0.24	8.7%	1.33	0%	1.20	0.24	11.1%	4.28	0%	1.21
Data targets	8–11%	4–6	25–40%			8–11%	4–6	25–40%		

Notes: All economies calibrated to quarterly repricing frequency $\bar{\Lambda} = 0.24$. “Freq” is the simulated frequency (confirming calibration). “ $E[|\Delta p|]$ ” is the mean absolute (non-sale) price change. “Kurt” is the raw kurtosis. “Sm%” is the fraction of price changes smaller than 2.5% in absolute value. $\kappa = (1+2\gamma)\bar{\Lambda}$ holds exactly when f_{ss} is supported in the interior of the inaction band and approximately to the extent that tail mass falls outside it; in the calibrated economies the approximation is tight under the standard normal-shock specification. The benchmark calibration is $\hat{\gamma} = 1$ via the hazard-exponent route; the kurtosis-route alternative $\hat{\gamma} \approx 0.4$ is reported for transparency. Data targets from Nakamura and Steinsson (2008), Midrigan (2011), and Alvarez, Le Bihan, and Lippi (2016). 15,000 firms, 150 quarters per simulation.

Table 2: Dominick’s price change moments (excl. sales)

Moment	Value
Frequency (weekly)	9.3%
Frequency (quarterly)	71.8%
Mean $ \Delta p $	13.5%
Kurtosis	7.62
Fraction $ \Delta p < 2.5\%$	21%
Fraction $ \Delta p < 5\%$	42%
N (price changes, excl. sales)	1,656,025

Notes: Sample moments from the Dominick’s Finer Foods scanner dataset (Chevalier, Kashyap, and Rossi, 2003), pooled across 7 product categories. Sale weeks are excluded by the algorithm of Midrigan (2011): a price change is flagged as a sale-related round-trip if the post-change price returns to its pre-change level within a window. Frequencies and price-change statistics are computed on the non-sale series. The quarterly frequency is computed as $1 - (1 - w)^{13}$ with w the weekly frequency. Kurtosis and the fractions of small price changes are non-sale and pooled.

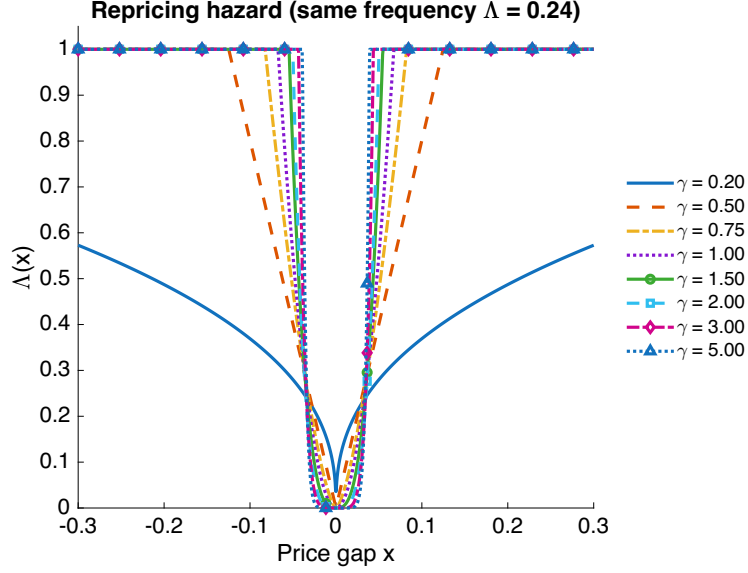


Figure 4: Repricing hazard across the spectrum

Notes: Repricing hazard $\Lambda(x)$ for selected values of γ . All economies calibrated to $\bar{\Lambda} = 0.24$ quarterly. Low γ produces a nearly flat hazard (close to Calvo); high γ produces a steep hazard (close to Golosov-Lucas).

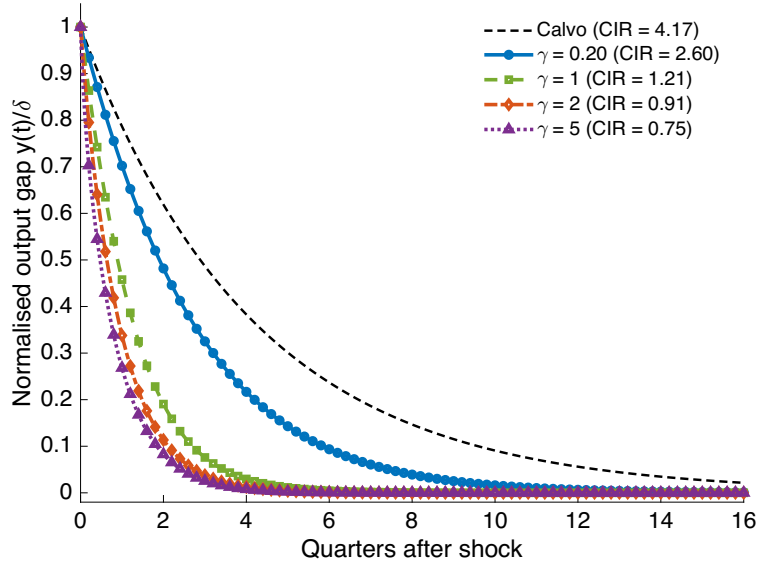


Figure 5: Output impulse response to a 1% monetary shock

Notes: All economies calibrated to $\bar{\Lambda} = 0.24$ quarterly; they differ only in γ . Higher γ implies stronger selection, faster price-level convergence, and smaller cumulative output effects. CIR values computed via [Alvarez, Le Bihan, and Lippi \(2016\)](#): $\text{CIR} = \text{Kurt}/(6 \cdot \text{Freq})$. The Calvo benchmark (dashed) has zero selection.

Table 3: Heterogeneous- σ_z moments at Dominick’s frequency

γ	$\bar{\xi}$	Kurtosis	Frac < 2.5%
0.10	0.0160	7.67	41%
0.15	0.0047	7.64	39%
0.20	0.0024	7.53	37%
0.25	0.0015	7.63	36%
0.30	0.0011	7.68	35%
0.40	0.0007	7.51	34%
Data	—	7.62	21%

Notes: Heterogeneous- σ_z simulation ($CV = 1$) with $\bar{\xi}$ calibrated for each γ to match $\bar{\Lambda} = 0.72$ quarterly. 8000 firms, 80 quarters after 40 quarters burn-in.

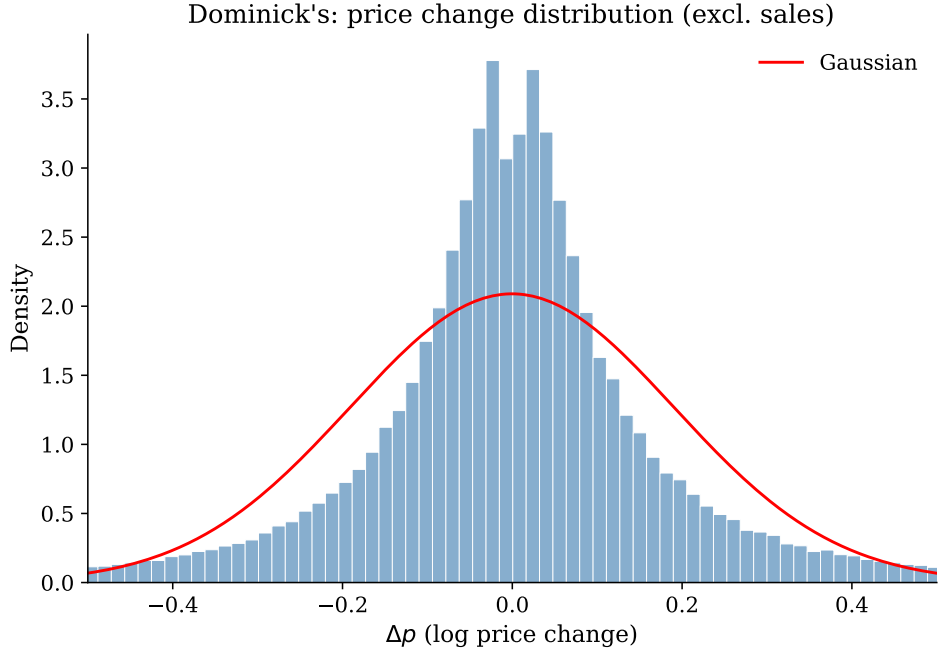


Figure 6: Distribution of non-sale price changes in Dominick’s

Notes: 7 categories, 42 million observations. The distribution is strongly leptokurtic (kurtosis = 7.6), with a sharp peak near zero and heavy tails relative to the Gaussian benchmark (red). Consistent with the model’s prediction for low γ .

F Comparison with Williams-Daly-Zachary

Section 2.2 of the main paper discusses the relationship between Theorem 1 and the Williams-Daly-Zachary (WDZ) theorem (McFadden, 1978, 1981). This appendix provides a systematic comparison. Both results share a common mathematical skeleton — an expected-maximum operator equals the Fenchel conjugate of a function determined by a distributional primitive

— but differ in their economic content, scope, and implications. Table 4 summarises the comparison along six dimensions.

Table 4: Williams-Daly-Zachary vs. Theorem 1

Dimension	WDZ	Theorem 1
Random element	Taste shocks ε_j on payoffs	Cost draws $w\xi$ on adjustment
Decision	Multi-alternative discrete choice	Binary: adjust or not
Surplus function	$E[\max_j \{u_j + \varepsilon_j\}]$	$\Pi^{FC}(S)$ (Definition 1)
Convexity result	Surplus convex in $(u_1, \dots, u_J)$	Payoff convex in S
Conjugate of	Generating function $\mathcal{G}(\varepsilon)$	$C(\varphi) = \int_0^\varphi G^{-1}(t) dt$
Choice probs	$p_j = \partial E[\max]/\partial u_j$	$\varphi^* = G(S)$
Primitive	Joint distribution of ε	Distribution G of costs
Coupling	Not constructed	Path-by-path equivalence
Economic context	Demand: consumer choice	Supply: adjustment frictions
Application	Multinomial logit (McFadden, 1974)	Calvo–GL spectrum

Notes: A side-by-side comparison of the Williams-Daly-Zachary (WDZ) theorem (McFadden, 1978, 1981) and Theorem 1 of the main paper. Both establish that an expected-maximum operator equals the Fenchel conjugate of a function determined by a distributional primitive; they differ in what plays the role of that primitive (taste shocks vs. adjustment costs), the economic mapping (demand vs. supply), and whether a path-by-path coupling is constructed. The binary case is formally equivalent under a sign change; the broader content of Theorem 1 ($G \mapsto C$ taxonomy, structural reading of Khan and Thomas (2008), the Calvo–Goloso-Lucas spectrum) is distinct from the WDZ literature.

For binary choice ($J = 2$), the two results are formally equivalent under a sign change: a random benefit $S - w\xi$ versus a random cost $w\xi$ applied to a fixed benefit S are isomorphic. The binary conjugate identity is therefore not new. The contributions of this paper lie beyond the duality identity itself: the mapping $G \mapsto C$ and the distributional taxonomy (Propositions 3–5), the structural explanation for the Khan and Thomas (2008) result, and the continuous parametrisation of the pricing spectrum are entirely absent from WDZ and the discrete choice literature.

G Computational routine

This appendix proves Theorem 2 and develops the dynamic FOC–envelope routine. The duality and the corresponding results extend to a multi-dimensional state $x \in \mathbb{R}^n$; for clarity I present the one-dimensional case. To match the dynamic pricing problem of the main text, the problem is stated in adjustment/non-adjustment branch form. Let $F^a(x', x; X)$ denote the per-period payoff when the agent adjusts the controlled state from x to x' , $F^n(x; X)$ the payoff when it does not, and $\delta \in [0, 1)$ a depreciation/decay parameter on the non-adjustment

branch.² The Bellman is

$$v(x; X) = \max_{p \in [0,1]} \left\{ p v^a(x; X) + (1-p) v^n(x; X) - C(p; X) \right\}, \quad (3)$$

$$v^a(x; X) = \max_{x'} \left\{ F^a(x', x; X) + \beta \mathbb{E} m(X, X') v(x'; X') \right\}, \quad (4)$$

$$v^n(x; X) = F^n(x; X) + \beta \mathbb{E} m(X, X') v(x(1-\delta); X'), \quad (5)$$

where v^a is the value of adjusting (optimised over the target x') and v^n the value of not adjusting; the extensive margin trades the two off subject to the convex cost C . Define the *adjustment surplus*

$$S(x; X) \equiv v^a(x; X) - v^n(x; X). \quad (6)$$

The surplus includes *both* the contemporaneous payoff difference $F^a(x'^*, x; X) - F^n(x; X)$ and the difference in continuation values from the two branches. The Fenchel duality (Theorem 1 of the main text) implies that the extensive margin in (3) admits the smooth probability-choice representation $\max_{p \in [0,1]} \{p S(x; X) - C(p; X)\}$, with interior first-order condition $C_p(p; X) = S(x; X)$. With the boundary projection,

$$p^*(x; X) = \begin{cases} 0, & S(x; X) \leq C_p(0; X), \\ (C_p(\cdot; X))^{-1}(S(x; X)), & C_p(0; X) < S(x; X) < C_p(1; X), \\ 1, & S(x; X) \geq C_p(1; X). \end{cases} \quad (7)$$

Equivalently, $p^*(x; X) = \min\{1, \max\{0, (C_p(\cdot; X))^{-1}(S(x; X))\}\}$. When C is induced by a fixed-cost distribution G , $C_p(p; X) = wG^{-1}(p)$ and the interior policy reduces to $p^* = G(S/w)$.

Proof of Theorem 2 (Value-function differentiability) The proof has three steps: (i) the extensive-margin maximisation has a smooth Fenchel envelope, so the post-extensive Bellman is a smooth maximisation over the intensive choice x' ; (ii) the operator has a unique fixed point in the space of bounded measurable functions by standard contraction-mapping arguments; (iii) the fixed point is C^1 in x by Santos (1991), Theorem 4.1, whose hypotheses are verified for the post-Fenchel Bellman under (A1)–(A3).

Step (i): Fenchel envelope. Under (A1), Proposition 1 gives $C(\cdot; X) \in C^2$ and strictly convex on $(0, 1)$. The static maximand

$$c^*(S; X) \equiv \max_{p \in [0,1]} \{p S - C(p; X)\}$$

²In the pricing application of the main text, $F^a(x', x; X) = R(x'; X)$ (post-adjustment profit, independent of x), $F^n(x; X) = R(x; X)$, and the non-adjustment state evolves by $x \mapsto x - \pi'$ (captured by an appropriate relabelling of $x(1-\delta)$). In a capital-adjustment application, F^a may depend on both x and x' via investment costs, and δ is the depreciation rate.

is therefore convex and C^1 in S , with $c^*(S; X) = p^*$ given by (7). Substituting into (3) and using (6) yields the post-Fenchel Bellman

$$v(x; X) = v^n(x; X) + c^*(S(x; X); X), \quad (8)$$

in which the extensive choice has been concentrated out into a smooth function of S .

Step (ii): contraction and fixed point. Define the operator T by the right-hand sides of (3)–(5). Under (A2), F^a and F^n are bounded C^1 functions; $C(\cdot; X)$ is bounded above on $[0, 1]$. By Blackwell’s sufficient conditions (Stokey, Lucas, and Prescott 1989, Thm. 3.3), T is a β -contraction on the Banach space $\mathcal{B}$ of bounded measurable functions $v(\cdot; X)$ under the sup-norm $\|v\|_0 \equiv \sup_{x, X} |v(x; X)|$. The unique fixed point $v^* \in \mathcal{B}$ therefore exists.

Step (iii): C^1 regularity via Santos (1991). C^1 regularity of the fixed point is shown by verifying that the operator T preserves the class of bounded C^1 functions, then applying the contraction in step (ii). Take a candidate continuation $w(\cdot; X')$ that is bounded and C^1 in its first argument. The adjustment branch

$$T^a(w)(x; X) \equiv \max_{x'} \{ F^a(x', x; X) + \beta \mathbb{E} m(X, X') w(x'; X') \}$$

is a smooth dynamic-programming maximisation. The relevant hypotheses must be stated on the *full maximand*, not on F^a alone. Strict concavity of F^a in x' does not by itself guarantee a unique smooth maximiser: the continuation $\beta \mathbb{E} m w(x'; X')$ may in principle break concavity or create multiple maximisers. (A3) therefore states the requirement directly on the full post-Fenchel maximand: it is uniformly strictly concave in x' , with a unique interior optimum and a non-singular SOC across (x, X) along the iteration. This is the joint hypothesis required by Santos (1991), Theorem 4.1. Under it, the theorem applies directly and yields $T^a(w)(\cdot; X) \in C^1$ in x , with derivative given by the Benveniste-Scheinkman envelope $\partial_x T^a(w) = F_x^a(x^*, x; X)$. (In the pricing case $F_x^a = 0$, recovering the main-text formula.) The substantive content of (A3) is bounding the discount-factor-weighted continuation curvature uniformly below F^a ’s primitive concavity, so the SOC survives along the iteration. The non-adjustment branch

$$T^n(w)(x; X) = F^n(x; X) + \beta \mathbb{E} m(X, X') w(x(1 - \delta); X')$$

is C^1 in x by direct differentiation under (A2) and $w \in C^1$. Composition with the smooth $c^*(\cdot; X)$ from step (i) gives the post-Fenchel iterate

$$T(w)(x; X) = T^n(w)(x; X) + c^*(T^a(w)(x; X) - T^n(w)(x; X); X) \in C^1.$$

So T maps the bounded C^1 class into itself. By the standard argument in Stokey, Lucas, and Prescott (1989, Ch. 3 and Lemma 6), the unique fixed point of step (ii) is in this class, i.e. $v^*(\cdot; X) \in C^1$ for each X , with v_x given by the branch-by-branch Benveniste-Scheinkman envelope stated below.

A clarification on the role of the duality at step (iii). With continuous G , the ex-ante integrated payoff $\Pi^{FC}(S) = \int \max\{S - w\xi, 0\} dG(\xi)$ already equals the smooth Fenchel conjugate $c^*(S)$ of C , so the ex-ante fixed-cost formulation is itself smooth in S . The duality's role is to make this smoothness manifest as a closed-form FOC $S = C'(\varphi)$ rather than an implicit integration over ξ , and equivalently to replace the discontinuous realised adjustment indicator $\mathbf{1}[\xi \leq S/w]$ by the smooth ex-ante probability $\varphi^* = G(S/w)$. Methods that track the realised threshold rule directly (e.g., locating the Ss-band cutoff in x for each ξ) face a kink in the conditional value function at the threshold gap, which obstructs the FOC-envelope iteration used in step (iii); the closed-form c^* representation lets the iteration proceed in the smooth FOC representation throughout. The ex-ante value itself loses smoothness only in the degenerate- G (Golosov-Lucas) limit, where c^* acquires a corner at $S = w\bar{\xi}$. Differentiability of v in the aggregate state X is a separate question depending on the smoothness of the aggregate transition in X , and is not required for the FOC system below; in heterogeneous-agent settings where X is a cross-sectional distribution, that question is handled by the projection or sequence-space machinery of [Reiter \(2009\)](#) and [Auclert, Rognlie, and Straub \(2020\)](#).

Proof of Corollary 3 (Policy differentiability) The joint objective in (p, x') in (3)–(4) is *not* jointly concave: the coupling term $p \mathbb{E} m v(x'; X')$ is bilinear in (p, x') , so strict convexity of C in p and strict concavity of F^a in x' do not deliver joint concavity. The two margins must therefore be treated separately.

Intensive margin. The maximisation defining v^a in (4) is independent of p . By (A2) the integrand is C^1 in x' , and by Theorem 2, $v \in C^1$ in its first argument. Strict concavity of F^a in x' (A2) and the strict SOC (A3) deliver a unique interior maximiser $x'^*(x; X)$ via the first-order condition

$$F_{x'}^a(x'^*, x; X) + \beta \mathbb{E} m(X, X') v_x(x'^*; X') = 0. \quad (9)$$

The implicit function theorem applied to (9), using the non-singular SOC, upgrades continuity to $x'^* \in C^1(x; X)$.

Extensive margin. Given $v^a, v^n \in C^1$ (Theorem 2), the surplus $S = v^a - v^n$ is C^1 in x . The extensive choice solves the static problem $\max_{p \in [0, 1]} \{p S(x; X) - C(p; X)\}$, strictly concave in p alone by (A1). The unique maximiser p^* is given by (7), continuous on (x, X) , and C^1 on the interior set $\{(x, X) : p^*(x; X) \in (0, 1)\}$ by the inverse function theorem applied to $C_p(\cdot; X)$.

Envelope and the practical gain Combining the envelope conditions from each branch with the extensive margin, the value derivative is

$$\begin{aligned} v_x(x; X) &= p^* F_x^a(x'^*, x; X) \\ &+ (1 - p^*) [F_x^n(x; X) + \beta(1 - \delta) \mathbb{E} m(X, X') v_x(x(1 - \delta); X')], \end{aligned} \quad (10)$$

the standard Benveniste-Scheinkman form (Benveniste and Scheinkman, 1979) applied branch by branch; v_x is well-defined by Theorem 2. In the pricing application $F^a(x', x; X) = R(x'; X)$ does not depend on x , so $F_x^a = 0$ and (10) reduces to $v_x = (1 - p^*)v_x^n$ as in the main text. The practical gain of the Fenchel formulation is that, with $v \in C^1$ from Theorem 2 and x'^* recovered from the FOC-envelope pair (9)–(10), the dynamic fixed-point iteration proceeds without an individual-level grid search for x'^* at each iteration; the recursion is endogenous-grid-like (Carroll, 2006), with the FOC system playing the role of the standard EGM update.

Global nonlinear solution of the spectrum model The differentiability of Theorem 2 also supports global nonlinear solution. I solve the full dynamic NK spectrum model by the repeated transition method (RTM) of Lee (2026). RTM constructs continuation values by matching periods along a simulated equilibrium path. Matched periods share the exogenous state and have a similar value of a scalar summary of the endogenous (distributional) state. No law of motion is parameterised, and the solution is not linearised around steady state. The FOC-envelope structure above is implemented directly: each period traces both the value V and its gradient V_x ; the matched conditional expectations $\mathbb{E} m V'$ and $\mathbb{E} m V'_x$ deliver the surplus $S = V^a - V^n$ (hence the hazard $\Lambda = G(S/w)$) and the resetter’s first-order condition $R_x(x'^*) + \beta \mathbb{E} m V_x(x'^* - \pi'; X') = 0$, with V_x updated by the envelope $V_x = (1 - \Lambda)V_x^n$. No individual-level grid search for x'^* is performed, exactly as the routine prescribes. The smoothness of V in x from Theorem 2 keeps the matched continuations continuous on the gap grid; implementations that work with the realised threshold rule, or the degenerate- G (Golosov-Lucas) limit, produce kinks in the conditional value function that compromise the matching step.

The scalar summary is the lagged price-dispersion index

$$\Delta_{t-1} = \int (P_i/P)^{-\varepsilon} f_{t-1}(x) dx.$$

In the calibration (`./codes/rtm/`; three-state Tauchen discretisations of preference, TFP, and monetary shocks, $3^3 = 27$ exogenous states), the object the matching step interpolates is the value slab $V(x; X)$ itself. Within each of the 27 exogenous states, $V(x; X)$ is monotone decreasing in Δ_{t-1} (Δ_{t-1} measures misallocation, so higher dispersion lowers value); this within-exogenous-state monotonicity supports Δ_{t-1} as a valid scalar summary for matching in the sense of Lee (2026). Figure G.1 visualises the within-exogenous-state alignment.

The steady-state consistency check across γ and the state-dependent demand GIRF based on this routine are reported in the global-analyses subsection of the main text.

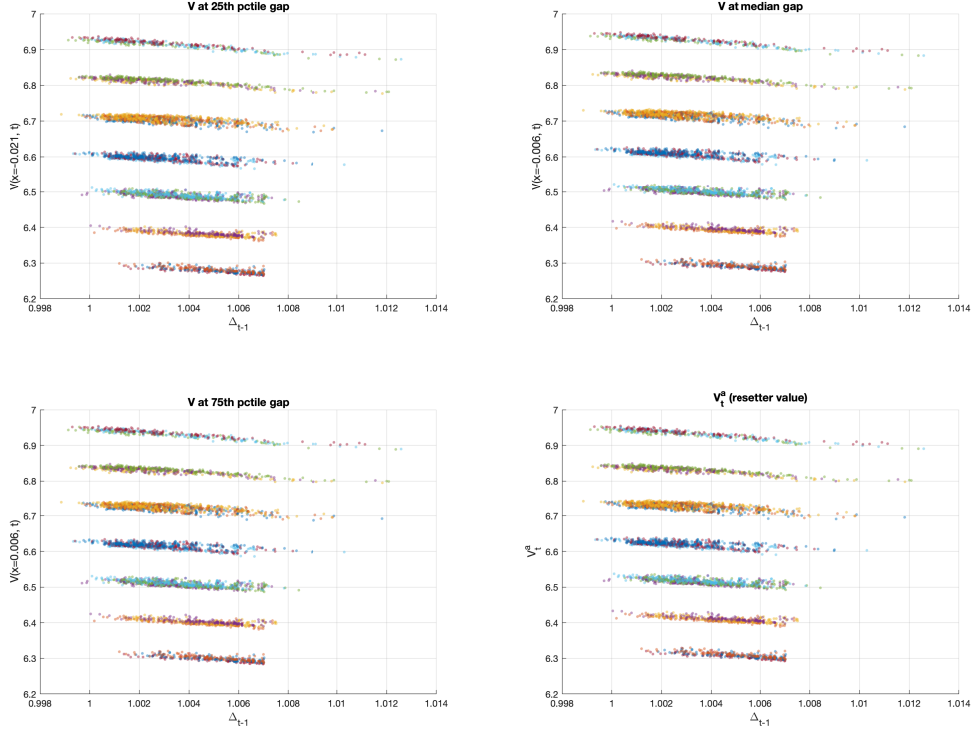


Figure 7: Within-exogenous-state monotonicity test: value $V(x; X)$ vs lagged price dispersion Δ_{t-1}

Notes: Each point is one simulated period t post burn-in; colour indicates the exogenous state (of $27 = 3^3$, from preference $\times$ TFP $\times$ monetary shocks). Panels (a)–(c) plot the value $V(x; X)$ at the 25th, 50th, and 75th mass percentiles of the steady-state gap distribution against the matching scalar Δ_{t-1} ; panel (d) is the resetter’s value V_t^a . Within each exogenous state the value is monotone in Δ_{t-1} — the within-exogenous-state monotonicity that the RTM matching exploits.

H Friction-extended duality and search applications

This appendix shows that the cost–distribution duality of Theorem 1 absorbs a class of general-equilibrium matching frictions without re-derivation: a scalar aggregate friction F multiplying the per-action surplus simply rescales the dual primitives. As a corollary of the construction, the canonical search-and-matching congestion externality and the [Hosios \(1990\)](#) efficiency condition reappear as recovered known results, not new claims of this paper; the dual machinery supplies a uniform representation, while the efficiency content is due to [Hosios \(1990\)](#) and, in non-Cobb–Douglas form, to [Mangin and Julien \(2021\)](#).

H.1 Friction-extended duality

Let $F > 0$ denote an aggregate friction multiplying the per-attempt surplus, so that an agent who chooses an adjustment rate $\varphi \in [0, 1]$ obtains operative surplus $\varphi \cdot (F S)$ net of the cost $C(\varphi)$ induced (via Theorem 1) by the idiosyncratic-cost distribution G . The agent solves

$$\max_{\varphi \in [0, 1]} \left\{ \varphi (F S) - C(\varphi) \right\},$$

with interior first-order condition

$$F S = C'(\varphi). \quad (11)$$

Equivalently, the random-cost representation underlying Theorem 1 prescribes adjustment whenever the idiosyncratic cost draw $\xi \sim G$ satisfies $\xi \leq F S$, so the implied adjustment rate is $\varphi = G(F S)$. Define the rescaled primitives

$$\tilde{G}(z) \equiv G(F z), \quad \tilde{C}(\varphi) \equiv \frac{1}{F} C(\varphi). \quad (12)$$

Then $\tilde{C}'(\varphi) = \frac{1}{F} C'(\varphi) = \frac{1}{F} G^{-1}(\varphi) = \tilde{G}^{-1}(\varphi)$, so (12) produces a dual cost–distribution pair in the sense of Theorem 1, and (11) is equivalently $S = \tilde{C}'(\varphi)$ with primitive $\tilde{G}$. Setting $F = 1$ recovers the baseline duality verbatim. The friction is therefore absorbed by a one-parameter rescaling of the dual pair; no re-derivation of the conjugate identity is required.

H.2 Canonical SaM instantiation

In the canonical search-and-matching environment of [Pissarides \(2000\)](#), the matching technology delivers a meeting probability $F = F(\theta)$ for the short side of the market, where $\theta \equiv v/u$ is market tightness (vacancies over searchers). Identifying F in (11) with this matching probability, the construction (12) provides the dual cost $\tilde{C}$ in the SaM environment as a direct relabelling of the cost–distribution pair obtained in the frictionless block. The random-cost / convex-cost equivalence of Theorem 1 carries over to the SaM environment with $F(\theta)$ supplied externally as a scalar aggregate friction; no separate derivation is needed.

H.3 Hosios efficiency condition as a recovered special case

When tightness depends on the aggregate adjustment rate, $\theta = \theta(\varphi)$, the matching friction $F = F(\theta(\varphi))$ inherits a dependence on φ that the atomistic agent in (11) takes as given. The planner instead solves

$$\max_{\varphi \in [0, 1]} \left\{ \varphi F(\theta(\varphi)) S - C(\varphi) \right\},$$

whose first-order condition reads

$$\underbrace{FS}_{\text{private}} + \underbrace{\frac{dF}{d\varphi} S \varphi}_{\text{congestion}} = \underbrace{C'(\varphi)}_{\text{planner}}. \quad (13)$$

Comparison with (11) identifies the wedge $-\frac{dF}{d\varphi} S \varphi$ as the canonical congestion externality of search theory: each marginal entrant alters the matching probability faced by all incumbents through θ . The Pigouvian entry tax

$$\tau^*(\varphi) \equiv -\frac{dF}{d\varphi} S \varphi$$

decentralises the planner’s allocation, because substituting $FS - \tau^*(\varphi) = C'(\varphi)$ into the agent’s problem reproduces (13).

Specialise to Cobb–Douglas matching $F(\theta) = A\theta^{1-\alpha}$ with matching elasticity $\alpha \in (0, 1)$, and let β denote the worker bargaining share that determines how surplus S is split. The wedge in (13) collapses to a condition relating β to α , and standard manipulation (Hosios 1990) shows that constrained efficiency obtains if and only if $\beta = \alpha$, the Hosios condition. The same algebra applied to a general matching technology $F(\theta)$ yields the bargaining-share / elasticity equality of Mangin and Julien (2021), under which the wedge again vanishes; the only change is that the elasticity α is replaced by the local elasticity of F at the equilibrium tightness.

Scope of the claim. The contribution here is representational: the dual cost–distribution construction of Theorem 1 absorbs GE matching frictions through the one-parameter rescaling (12), and the canonical efficiency results follow as a special case of the same algebra. The efficiency content recovered above is due to Hosios (1990) and Mangin and Julien (2021), respectively; the cost-shape parameter γ of the main text and the matching elasticity α remain orthogonal margins, governing the curvature of C and the externality wedge in (13) independently.

References

- Alvarez, Fernando, Hervé Le Bihan, and Francesco Lippi. 2016. “The Real Effects of Monetary Shocks in Sticky Price Models: A Sufficient Statistic Approach.” *American Economic Review* 106 (10):2817–2851.
- Alvarez, Fernando, Francesco Lippi, and Aleksei Oskolkov. 2022. “The Macroeconomics of Sticky Prices with Generalized Hazard Functions.” *The Quarterly Journal of Economics* 137 (2):989–1038.
- Auclert, Adrien, Matthew Rognlie, and Ludwig Straub. 2020. “Micro Jumps, Macro Humps:

- Monetary Policy and Business Cycles in an Estimated HANK Model.” Working Paper 26647, National Bureau of Economic Research.
- Benveniste, L. M. and J. A. Scheinkman. 1979. “On the Differentiability of the Value Function in Dynamic Models of Economics.” *Econometrica* 47 (3):727–732.
- Carroll, Christopher D. 2006. “The Method of Endogenous Gridpoints for Solving Dynamic Stochastic Optimization Problems.” *Economics Letters* 91 (3):312–320.
- Chevalier, Judith A., Anil K. Kashyap, and Peter E. Rossi. 2003. “Why Don’t Prices Rise During Periods of Peak Demand? Evidence from Scanner Data.” *American Economic Review* 93 (1):15–37.
- Gabaix, Xavier. 2014. “A Sparsity-Based Model of Bounded Rationality.” *Quarterly Journal of Economics* 129 (4):1661–1710.
- Hosios, Arthur J. 1990. “On the Efficiency of Matching and Related Models of Search and Unemployment.” *The Review of Economic Studies* 57 (2):279–298.
- Khan, Aubhik and Julia K. Thomas. 2008. “Idiosyncratic Shocks and the Role of Nonconvexities in Plant and Aggregate Investment Dynamics.” *Econometrica* 76 (2):395–436.
- Lee, Hanbaek. 2026. “A Global Dynamic Nonlinear Solution Framework and the Repeated Transition Method.” Working paper, University of Cambridge.
- Mangin, Sephorah and Benoît Julien. 2021. “Efficiency in Search and Matching Models: A Generalized Hosios Condition.” *Journal of Economic Theory* 193:105208.
- Matějka, Filip and Alisdair McKay. 2015. “Rational Inattention to Discrete Choices: A New Foundation for the Multinomial Logit Model.” *American Economic Review* 105 (1):272–298.
- McFadden, Daniel. 1974. “Conditional Logit Analysis of Qualitative Choice Behavior.” In *Frontiers in Econometrics*, edited by Paul Zarembka. Academic Press, 105–142.
- . 1978. “Modelling the Choice of Residential Location.” In *Spatial Interaction Theory and Planning Models*, edited by A. Karlqvist et al. North-Holland, 75–96.
- . 1981. “Econometric Models of Probabilistic Choice.” In *Structural Analysis of Discrete Data*, edited by Charles F. Manski and Daniel McFadden. MIT Press, 198–272.
- Midrigan, Virgiliu. 2011. “Menu Costs, Multiproduct Firms, and Aggregate Fluctuations.” *Econometrica* 79 (4):1139–1180.
- Nakamura, Emi and Jón Steinsson. 2008. “Five Facts about Prices: A Reevaluation of Menu Cost Models.” *Quarterly Journal of Economics* 123 (4):1415–1464.

- . 2010. “Monetary Non-Neutrality in a Multisector Menu Cost Model.” *Quarterly Journal of Economics* 125 (3):961–1013.
- Oskolkov, Aleksei and Francesco Lippi. 2026. “Misallocation with Lumpy Investment.” Discussion Paper 21212, Centre for Economic Policy Research.
- Pissarides, Christopher A. 2000. *Equilibrium Unemployment Theory*. MIT Press, 2nd ed.
- Reiter, Michael. 2009. “Solving Heterogeneous-Agent Models by Projection and Perturbation.” *Journal of Economic Dynamics and Control* 53 (6):649–665.
- Santos, Manuel S. 1991. “Smoothness of the Policy Function in Discrete Time Economic Models.” *Econometrica* 59 (5):1365–1382.
- Sims, Christopher A. 2003. “Implications of Rational Inattention.” *Journal of Monetary Economics* 50 (3):665–690.
- Stokey, Nancy L., Robert E. Lucas, and Edward C. Prescott. 1989. *Recursive Methods in Economic Dynamics*. Cambridge, MA: Harvard University Press.